

100 Days of Azure DevOps

Day 36 Handout — A percentage is not a canary. A watch plus a named abort is. Averages hide a dying bird

4 - Continuous Delivery · Personal lab only · Educational content · Not a sales pitch

Architecture A — three ways to expose a small risk

Mechanism	What moves	Azure lab knob
Slot traffic %	Fraction of requests to staging slot	App Service Testing in production (Standard+)
Feature flag	Code path on/off per user/cohort	App Configuration / your own flag — no fleet split
Revision weight	Later: Container Apps / API Management	Not required today
Abort	0% / flag off / swap back	Must be named before the ramp starts
Watch	5xx, latency, a business heartbeat on the canary only	Do not watch the blended average

Architecture B — ramp vs blue-green vs flag

Goal	Choose	Skip
Prove on 5% of real users	Canary traffic split + isolated metrics	Blue-green 100% flip
Hide a path with no split	Feature flag	Calling a flag a canary without traffic
Instant all-or-nothing rollback	Blue-green swap (Day 35)	A 40-minute ramp you cannot stop
No Standard SKU	Paper plan + abort metric — still the lab	Claiming Testing-in-prod on F1

Architecture C — pitfalls that lunch through the watch

Pitfall	What you see	Fix
Blended dashboard	Error rate 'fine' while 5% is on fire	Filter by slot / revision / flag
No abort signal	Team debates in chat at 12%	Write: 5xx > N for 15 min → 0%
5% then lunch	Unattended experiment on customers	15 min watch is a calendar block, not a vibe
Flag + split, no diagram	Nobody knows who saw what	Pick one mechanism for the paper plan

One-liner to remember

Canaries in coal mines and canaries in prod share a job: die early so the rest of us don't — write the abort before the ramp.

Step-by-step lab (personal Azure / Azure DevOps only)

Personal Azure only. Paper plan is mandatory. Testing-in-production is optional and SKU-gated.

- 1 Write docs/canary-plan-day36.md for your webapp: 5% → watch 15 min → 25% → watch → 100% or abort.
- 2 Name the abort metric in that file (example: 5xx rate on the canary slot, or a failed heartbeat). If you cannot name it, you do not have a canary.
- 3 Portal → Web app → Deployment slots → Testing in production (if present). Note whether routing % exists on your SKU.
- 4 If routing exists: send 5% to staging for a few minutes, then 0%. Do not leave a split overnight.
- 5 Add one sentence: flag vs traffic split — which one this lab would use and why (not both).
- 6 State that blended averages are forbidden as the watch. Isolated canary metrics only.

Canary checklist (paper is the product; CLI is optional)

```
# docs/canary-plan-day36.md
# Ramp: 5% (15 min) → 25% (15 min) → 100% or abort
# Abort: canary-slot 5xx > 2% for 5 min → set routing to 0% (or swap back)
# Watch: slot-filtered metrics, not the app-wide average
# Mechanism this lab: traffic % on staging | feature flag (pick one)
#
# Optional (Standard+): Portal → Testing in production → route % to staging
# az webapp traffic-routing set -g <rg> -n <app> --distribution staging=5
```

Done checklist

- Abort metric is written, not 'we will see'
- Ramp times are on the page
- Know Testing-in-prod is SKU-gated
- Did not leave a traffic split running

Tomorrow — Day 37

Rolling deployments — change tires while moving; N and N-1 must coexist or you scheduled a partial outage.